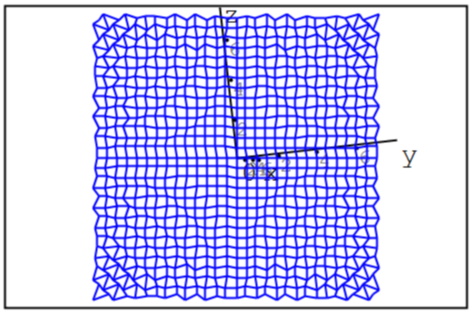
peter.vlasschaert@gmail.com,13/02/2022
simple 3D graphics (Smath & Maxima)

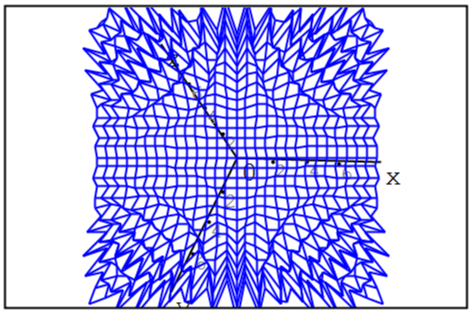
$f(x; y) := \sin(x \cdot y)$



$f(x; y)$

$a := \left(\left(\mathrm{Diff}(f(x; y); x)\right)\right)$

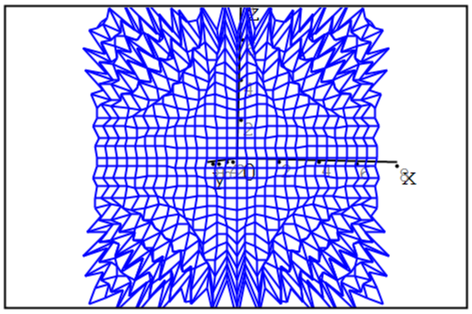
$a = y \cdot \cos(x \cdot y)$



a

$g(x; y) := a$

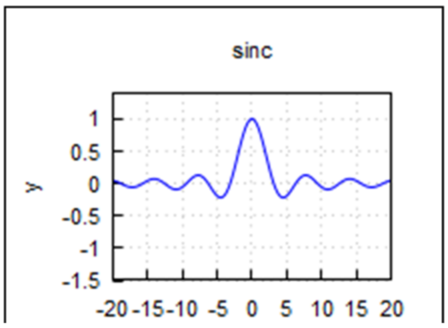
$g(x; y) = y \cdot \cos(x \cdot y)$

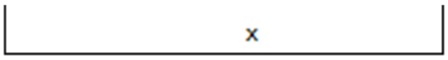


$g(x; y)$

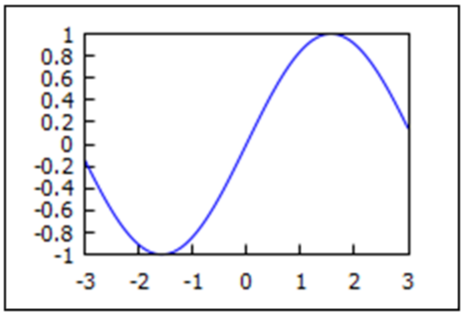
Plotting with Maxima

2D plotting





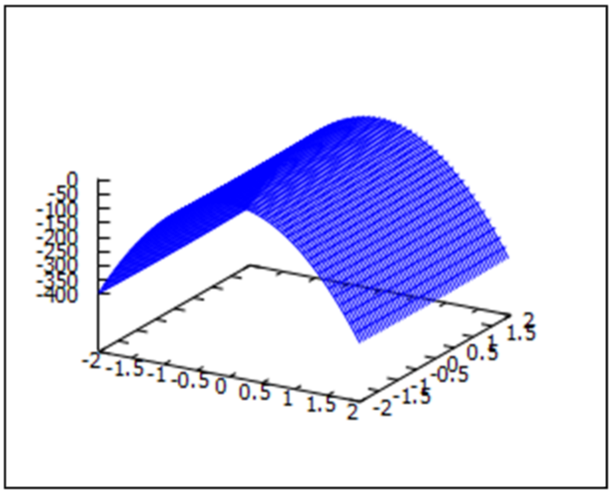
$\left\{ \mathit{explicit}\left(\frac{\sin(x)}{x};\ x;\ -20;\ 20 \right) \right.$



$\mathrm{Draw2D}\left(\mathit{explicit}\left(\sin(x);\ x;\ -3;\ 3 \right);\ \begin{cases} 6\ \mathrm{cm} \\ 4\ \mathrm{cm} \end{cases} \right)$

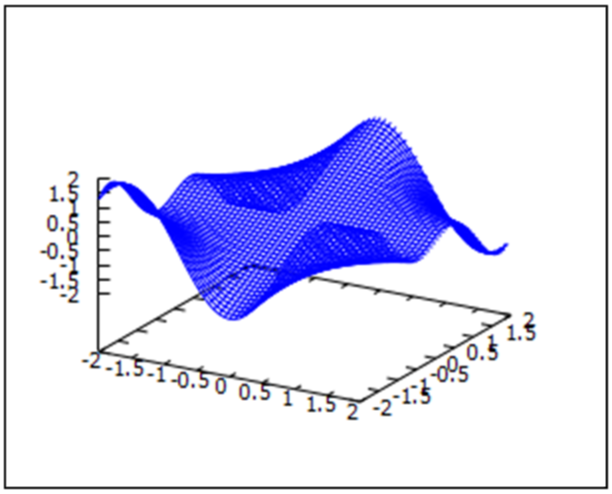
3D plotting

$f1 := \mathrm{Draw3D}\left(\mathit{explicit}\left(y^2 - (10 \cdot x)^2;\ x;\ -2;\ 2;\ y;\ -2;\ 2 \right) \right)$



$f1$

$f2 := \mathrm{Draw3D}\left(\mathit{explicit}\left(g(x;\ y);\ x;\ -2;\ 2;\ y;\ -2;\ 2 \right) \right)$



$f2$